

Credit covariance and routing in dendritic trees

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Abstract

Biological learning must transform a small number of cellular teaching signals into distinct updates at many synapses. We formulate this as a constrained routing problem. In a directed conductance-based dendritic tree, each exact synaptic gradient separates into a synapse-local eligibility and a somatic error transported through the unique path to that synapse. A feedback circuit therefore chooses both a coordinate, such as neuron identity, and a routing subspace over compartments. We define credit capture as the gradient energy in that subspace and show that it determines a one-step descent guarantee. Across a task distribution, the relevant object is the covariance of exact credit, not feedback rank alone: the optimal unconstrained channels are covariance eigenvectors, whereas biological channels are restricted to ancestry-supported routes. Conductance regulates the gain of those routes, and focal inhibition attenuates descendants selectively at fixed state. The theory predicts that learning succeeds when task-credit covariance aligns with the anatomical route dictionary and that route overlap controls interference. These results define a communication-level theory linking local plasticity, dendritic topology and task structure.

1 Introduction

Credit assignment asks how an outcome-dependent teaching signal can produce the parameter-specific changes required for learning. Backpropagation solves this problem by communicating one derivative to every parameter [1]. Biological synapses instead update from signals available locally, motivating three-factor plasticity, feedback alignment and dendritic learning theories [2–4]. Most formulations specify how a neuron receives an error but do not specify how that error is addressed across a branched dendritic tree.

We distinguish three objects. A *teaching coordinate* identifies which neuron or functional population should change. A *route* determines which compartments within that neuron share the coordinate. A synapse-local *eligibility* converts the routed signal into a conductance update. This separation makes feedback bandwidth, dendritic topology and conductance comparable within one formalism.

The theory has four parts. First, exact differentiation of a directed conductance tree identifies the information required at each synapse. Second, we treat restricted feedback as projection into a route subspace and relate projection energy to descent. Third, we replace a single gradient by its task distribution and obtain a covariance-matching objective. Finally, we derive consequences for inhibitory route gain and task interference. The claims concern communication and local learning; they do not assert that a particular biological circuit computes oracle projection coefficients.

2 Exact credit in a conductance tree

Consider a directed rooted tree. Compartment n receives synapses $i \in \mathcal{S}_n$ and child compartments $c \in \mathcal{C}_n$. Its steady-state voltage is

$$V_n = R_n^{\text{tot}} \left(\sum_{i \in \mathcal{S}_n} g_i x_i E_i + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c + g_n^L E_L \right), \quad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}, \quad (1)$$

where $a_c = f_c(V_c)$ and total conductance includes leak, active synaptic conductance and child coupling.

Theorem 1 (Local eligibility and transported error). *For a loss $\mathcal{L}$ that depends on the root output, the gradient of a synaptic conductance g_i at compartment n is*

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}}(E_i - V_n)}_{e_i, \text{ local eligibility}} \underbrace{\frac{\partial \mathcal{L}}{\partial V_n}}_{\delta_n, \text{ compartment error}}. \quad (2)$$

If neuron u has somatic error $\delta_{0,u}$, then

$$\delta_n = \delta_{0,u} \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(j \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_j(V_j) R_k^{\text{tot}} g_{j \rightarrow k}^{\text{den}}. \quad (3)$$

Equation 2 follows by differentiating the numerator and denominator of Eq. 1; the driving-force term $E_i - V_n$ is the conductance-specific contribution. Equation 3 follows by iterating the child-to-parent derivative along the unique path. Thus an exact gradient requires a local eligibility, a neuron-specific somatic coordinate and path-dependent transport. A global scalar broadcast removes the second and third objects simultaneously. Neuron-indexed feedback restores the coordinate; exact local path factors would then restore the remaining compartment field.

Numerical reconstruction of Eqs. 2–3 matches automatic differentiation to below 2×10^{-7} maximum relative error across the archived diagnostic cohort (Fig. 1a). In regular $[3, 3]$ trees, preserving neuron identity raises the cosine between available and exact dendritic gradients from approximately zero to 0.708 in conductance-based trees and 0.625 in matched additive trees (Fig. 1b,c). Exact path transport makes the theorem-derived eligibility sufficient to match the same-architecture backpropagation reference (Fig. 1d).

3 Feedback as a route subspace

Let $A_T \in \{0, 1\}^{p \times q}$ be the synapse-by-route ancestry matrix of tree T : $(A_T)_{ik} = 1$ when route site k lies between synapse i and the soma. With local or anatomical weights W_e , route gains β and a channel assignment C , an m -channel feedback vector $\mathbf{z}$ generates

$$\hat{\mathbf{g}} = D_T \mathbf{z}, \quad D_T = W_e A_T \text{diag}(\beta) C. \quad (4)$$

Topology fixes support. Conductance changes gains and conditioning but cannot create an ancestry relation absent from A_T .

Let $W \succ 0$ be diagonal, $\bar{\mathbf{g}} = W^{1/2} \mathbf{g}$ and $\bar{D} = W^{1/2} D_T$. Denote by P_D the Euclidean orthogonal projector onto $\text{col}(\bar{D})$. Define

$$R(T, \mathbf{g}; W) = \frac{\|P_D \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2}. \quad (5)$$

Theorem 2 (Capture and one-step descent). *Let $\bar{\mathcal{L}}$ have L_W -Lipschitz gradient in the transformed coordinates. For $0 < \eta < 2/L_W$,*

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta P_D \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \left(1 - \frac{L_W \eta}{2}\right) \|P_D \bar{\mathbf{g}}\|_2^2. \quad (6)$$

At $\eta = 1/L_W$, the projected-to-full-gradient descent guarantees have ratio $R(T, \mathbf{g}; W)$.

The proof is the descent lemma followed by $\bar{\mathbf{g}}^T P_D \bar{\mathbf{g}} = \|P_D \bar{\mathbf{g}}\|_2^2$. Ten thousand random positive-definite quadratic trials produced no bound violations; the minimum numerical slack was 2.67×10^{-7} and the capture identity agreed to 1.89×10^{-15} . The theorem is a one-step statement for an exact gradient expressed in declared coordinates. It does not show that a circuit can infer the least-squares coefficients $\mathbf{z}$.

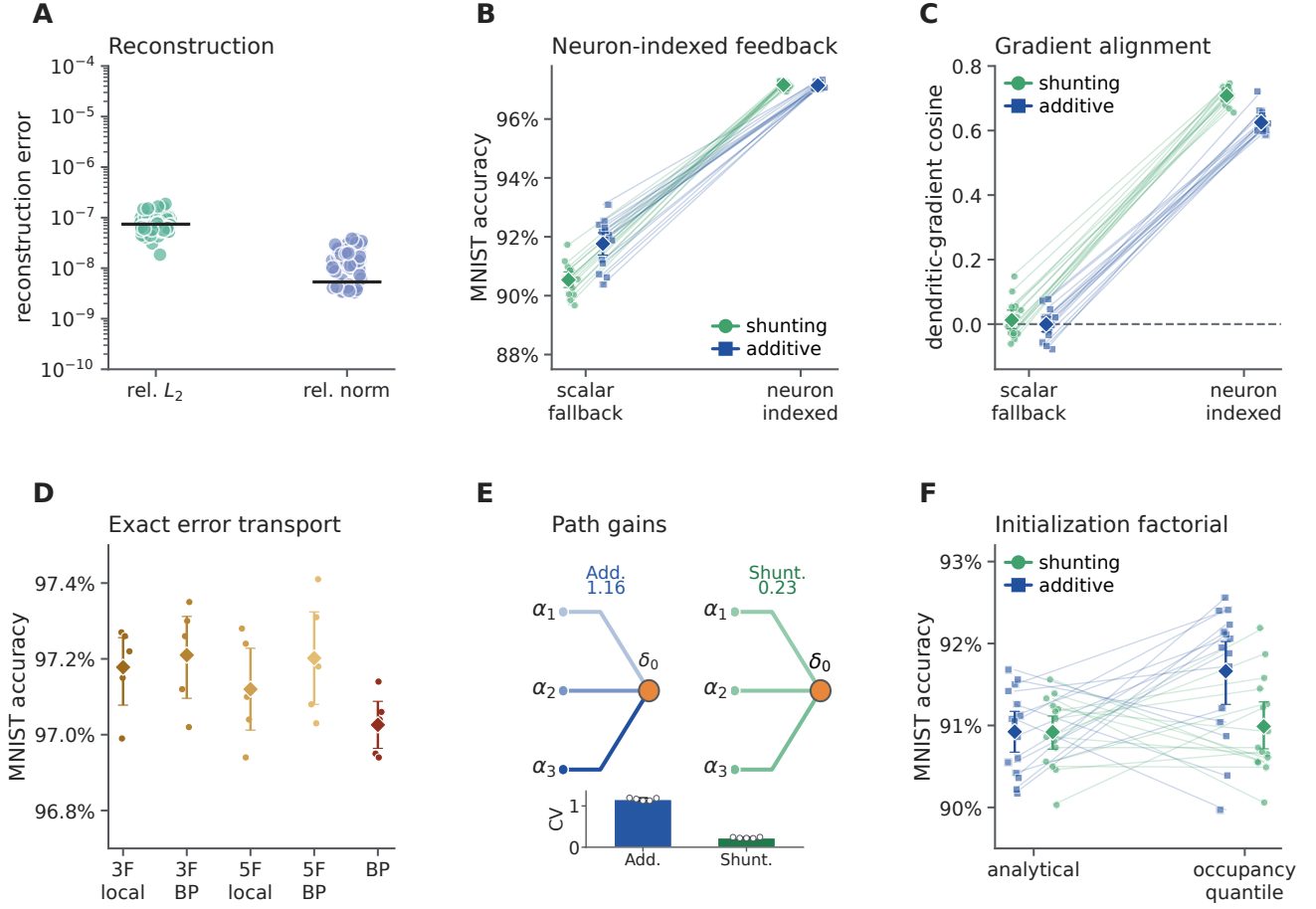


Figure 1: **The principal feedback limitation is loss of teaching-signal identity.** Exact analytic gradients agree with automatic differentiation (A). Neuron-indexed feedback improves both learning and gradient direction relative to scalar fallback (B,C). Exact transport isolates the local eligibility (D); the remaining panels separate conductance operating point, initialization and one-step optimization. This figure is shared with the comprehensive working draft and would be redrawn or assigned to only one paper before independent submission.

4 Task distributions select useful routes

A single gradient cannot determine whether a route dictionary will be useful over learning. Let $C_g = \mathbb{E}[\bar{\mathbf{g}}\bar{\mathbf{g}}^T]$ be the uncentered covariance of exact task credit. We define energy-weighted expected capture as the ratio of expected projected and total energy.

Theorem 3 (Covariance matching). *For a fixed route subspace,*

$$R(T, C_g) = \frac{\text{tr}(P_D C_g)}{\text{tr}(C_g)}. \quad (7)$$

Corollary 1 (Unconstrained upper bound). *Among all m -dimensional subspaces, Eq. 7 is maximized by the leading m eigenvectors of C_g . The maximum is $\sum_{j=1}^m \lambda_j(C_g) / \text{tr}(C_g)$.*

The corollary is the Ky Fan maximum principle. It separates channel count from basis choice. A rank- m PCA dictionary is an unconstrained information oracle; an ancestry dictionary is restricted to nested biological supports. The relevant comparison is therefore not whether exact credit is low rank, but how much task covariance lies in the route subspace available to the cell.

This condition explains both success and failure. When exact gradients are constructed with a fraction a of their energy in a fixed eight-route MICrONS subspace, measured capture equals a and predicts one-step and iterative loss progress (Fig. 2). When gradients are derived from a small measured visual-response cohort without imposing alignment, morphology does not reliably exceed ancestry-shuffled routes. Anatomical capacity is therefore not evidence of task alignment.

5 Conductance provides descendant-selective gain control

Let α_n^{cond} be Eq. 3 without reactivation derivatives. At fixed forward state and fixed coupling conductances, add inhibitory conductance $\Delta G_k^I \geq 0$.

Proposition 1 (Fixed-state shunting selectivity).

$$\frac{\alpha_n^{\text{cond}}(\Delta G^I)}{\alpha_n^{\text{cond}}(0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}}{g_k^{\text{tot}} + \Delta G_k^I}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n), \end{cases} \quad (8)$$

where $\mathcal{A}(n)$ is the path from n to the soma.

Thus a focal conductance change attenuates a nested descendant domain. The result does not imply that inhibition improves learning: if $z_n = \log \alpha_n$ and the attenuation is β_n , then $\text{Var}(z - \beta) = \text{Var}(z) + \text{Var}(\beta) - 2\text{Cov}(z, \beta)$. Gain dispersion narrows only when the attenuation covaries sufficiently with the original gain. Shunting is a route-gain operator, not a universal gradient preconditioner.

6 Route overlap predicts interference

Let Task A be at a local quadratic optimum with Hessian $H_A \succeq 0$. A Task-B update routed by M_B is $-\eta M_B \mathbf{g}_B$. Exact expansion gives

$$\Delta \mathcal{L}_A = \frac{\eta^2}{2} \mathbf{g}_B^T M_B^T H_A M_B \mathbf{g}_B. \quad (9)$$

For equal-size routes with fractional overlap ω , opposing requested updates on shared coordinates and off-route leakage λ , the normalized loss increase is

$$\Delta \mathcal{L}_A = \frac{\eta^2}{2} [4\omega + \lambda^2(1 - \omega)]. \quad (10)$$

The identity makes two independent failure modes explicit: reuse of the same route by incompatible tasks and leakage outside the requested route. Numerical evaluation agrees with explicit vector updates to 4.16×10^{-17} maximum absolute error (Fig. 3). It is static and does not model spiking, eligibility timing or consolidation.

7 Discussion

The main result is a change in the level at which local credit assignment is posed. A cellular teaching signal is not merely too small or noisy. It must be addressed. Neuron identity supplies a coarse coordinate; dendritic ancestry expands it into nested supports; local eligibility resolves synapse-specific sign and scale; conductance regulates route gain. This decomposition connects three-factor plasticity to an explicit communication problem.

The covariance formulation prevents two common overinterpretations. First, low rank is not sufficient: an unconstrained singular vector need not be implementable by a dendritic route. Second, anatomical route capacity is not task relevance. The same tree can be well suited to one credit covariance and poorly aligned with another. Negative task-alignment results are therefore a boundary of the theory rather than a contradiction.

The framework yields experimentally separable predictions. Teaching signals should preserve neuron identity before resolving within-tree location. Focal conductance changes should alter plasticity preferentially in descendant domains when somatic output is controlled. Tasks that recruit overlapping routes should interfere more strongly. Testing these predictions requires longitudinal, branch-resolved learning data; the present mathematics alone does not identify a biological implementation.

8 Methods and numerical checks

All projections use an explicit weighted coordinate transformation before ordinary Euclidean least squares. Capture is the ratio of projected to total energy, not the mean of unweighted sample-wise cosines. The descent theorem was verified on 10,000 random positive-definite quadratics at $\eta = 1/L_W$. The interference surface used 101 values each of ω and λ and was checked against explicit vector updates. Regular-tree numerical diagnostics, controlled alignment experiments and route-interference controls use the frozen tables and code documented in the comprehensive project repository.

Data and code availability

This working companion draft uses only derived source tables and scripts in the parent journal project. A separate archival release and a non-overlapping figure set would be prepared before submission.

Competing interests

The authors declare no competing interests.

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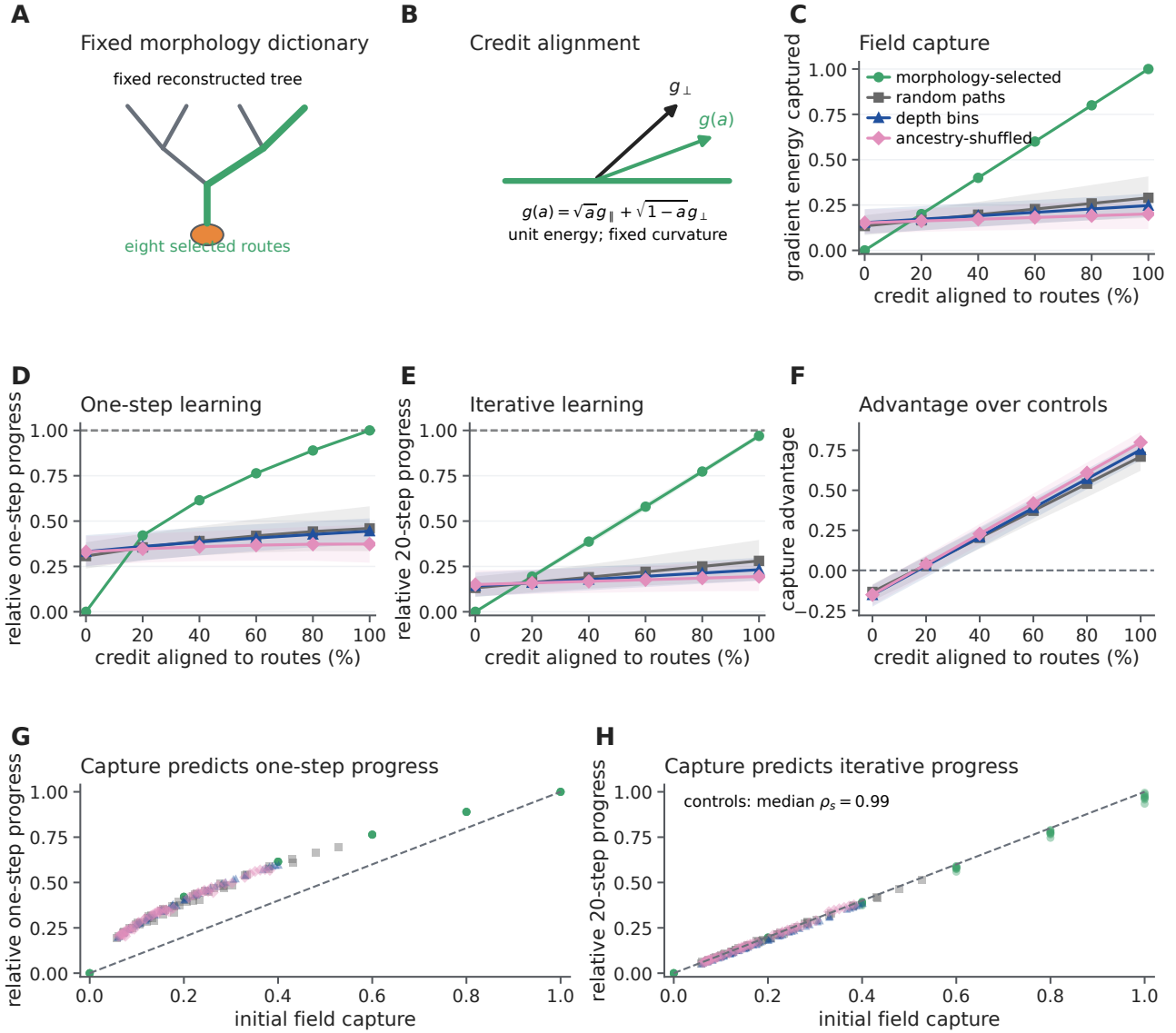


Figure 2: Route value is conditional on task-credit alignment. Controlled gradients interpolate between the route span and its orthogonal complement at fixed total energy. Capture and optimization progress rise with imposed alignment, whereas structural controls do not receive this advantage. This controlled construction establishes sufficiency, not biological use of the oracle coefficients.

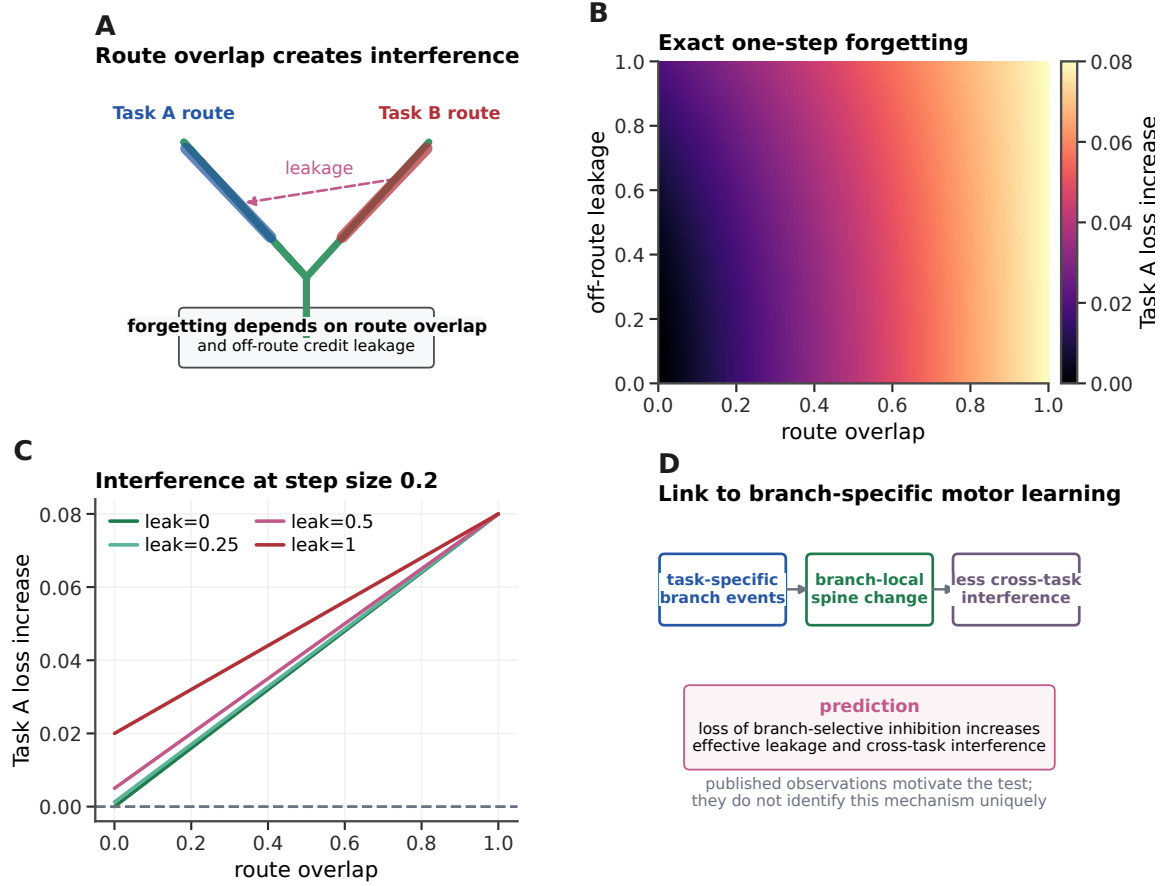


Figure 3: Overlap and off-route leakage determine one-step interference. The heat map evaluates Eq. 10; line slices show how anatomical route reuse and imperfect feedback make separable contributions. The biological link is a prediction, not a causal test.